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TYCHO BRAHE'S ATTITUDE TOWARDS ASTROLOGY AND HIS RELATIONS TO HEINRICH RANTZAU

Summary

Contrary to his astronomical works and observations the astrological activities of Tycho Brahe have received less attention and are usually treated as an aberration characteristic for an astronomer of the Early Modern Period. The paper deals with Tycho's astrological commissions for the Danish court, his attempts to create a reformed astrology and his relations to Heinrich Rantzau (1526–1598), who was himself an ardent believer in astrology.

Both Tycho Brahe and Heinrich Rantzau (1526–1598), Danish governor of Schleswig and Holstein, were of old and noble lineage. With his passion for books, his patronage in various fields of humanistic learning and art Rantzau was considered a black swan among contemporary noblemen. His father Johann, who was an influential politician and commander became governor of Schleswig-Holstein and witnessed Luther's appearance at the Imperial Diet at Worms.¹ Deeply impressed he joined the Protestant camp and later on sent his son to study in Wittenberg. During his long stay for about eight years young Heinrich was influenced by Philipp Melanchthon, who was an ardent believer in astrology. Melanchthon had studied in Tübingen under the eminent astronomer Johannes Stoeffler (1452–1531) and Rantzau was but one of his numerous students who left Wittenberg with a staunch belief in the new faith and the old art of astrology.²

1 Michael Beuther, *Kurtzbegriffene Anzeygung/ vom Leben/ Stannde und Wesen/ der Gestrengen/ Edlen und Vesten/ Herrn Johann Ranzawen Ritters/ trejer Könige zu Dänemarck/ etc. gewesenene FeldObersten: Hainrich Ranzawen/ seines Sohns/ Königlicher Mayestet zu Dänemarck noch jetziger zeit verordneten Statthalters/ in den Fürstenthummen Sleßwick/ Holsaten/ Dietmarsen/ etc.* [...], Basel 1582, p. 368.

2 For a more detailed account see the study *Heinrich Rantzau und die Astrologie: Ein Beitrag zur Kulturgeschichte des 16. Jahrhunderts* by the author (Postdoctoral thesis, Hamburg 2001; forthcoming as vol. 2 of *Disquisitiones Historiae Scientiarum: Braunschweiger Beiträge zur Wissenschaftsgeschichte* [Braunschweig 2004]).

After taking over the ancestral seat of Breitenburg near Hamburg in 1563 Heinrich Rantzau established a library, which finally contained 6300 volumes in the main fields of knowledge: Logic, Mathematics, Physics, Medicine, Law, Theology and History. The end of this magnificent collection came in 1627, when Wallenstein's troops besieged Breitenburg and caused a havoc. Major parts of Rantzau's library were carried off to Prague and later on fell into the hands of Swedish troops.³ Thus single volumes have been dispersed all over Northern Europe, but a considerable number of Count Rantzau's books still can be found in the National and University Library at Prague.⁴

Even these remains preserved by chance indicate a very well assorted library in the field of astronomy and astrology. The Danish governor must have possessed almost every relevant astrological publication of the sixteenth century, not to speak of manuscripts and numerous scientific instruments.⁵ Rantzau's collection formed the starting point for his own publications, for he was himself an ambitious and prolific writer of several textbooks covering all aspects of the art. The fact that Rantzau's books are mainly industrious compilations of other works (including manuscripts) is in itself significant: In astrology not originality but referring to established authors was demanded with the intention to advance the art by understanding it properly.

The astronomical activities on the island of Hven did not escape the attentiveness of the Danish governor. Because also Tycho busied himself with astrological occupations one would expect to find traces in the correspondence of the two noblemen, but the 13 letters extant of the period between 1585–1597 mainly deal with the problem of paper supply for Tycho's printing press which he hoped to get from Rantzau's paper mill.⁶ In his first letter from 1585 Tycho also asked to get on loan a catalog of astrological books preserved in the governor's famous library in

3 Marcus Posselt, „Die Bibliothek Heinrich Rantzau's“, *Zeitschrift der Gesellschaft für Schleswig-Holstein-Lauenburgische Geschichte*, 11, 1881, p. 69–124.

4 Isak Collijn, „Rester av Heinrich Rantzaus bibliotek på Breitenburg i National- och Universitetsbiblioteket i Prag“, *Nordisk Tidskrift för Bok- och Biblioteksväsen*, 26, 1939, p. 125–153; 27, 1940, p. 179–238; 28, 1941, p. 1–14.

5 Peter Lindeberg, *Hypotyposis Arcium, Palatiorum, Librorum, Pyramidum, Obeliscorum, Cipporum, Molarum, Fontium, Monumentorum et Epitaphiorum, ab illustri et strenuo Viro Henrico Ranzovio, Prorege et Equite Holsato, conditorum [...]*, Frankfurt/M. 1592, p. 23.

6 TBOO, vol. 7, p. 89f. (Hven, 8.1.1585).

the castle of Breitenburg.⁷ Three years later Tycho sent a horizontal sundial (*Sciotericum horizontale*) to Heinrich Rantzau and thanked for his efforts to supply him with paper.⁸ Shortly afterwards Rantzau had confirmed the receipt of the instrument and also of a device suitable to explain the Tychonic system of the world easily.⁹ In 1589 Tycho Brahe set up a paper mill on Hven¹⁰ and asked Heinrich Rantzau to get a paper-maker (*Chartopaeus*) for him.¹¹ The last letters from 1597 deal with Tycho's wish to settle down somewhere on Rantzau's estate after he had fallen out of favour with King Christian.¹²

Heinrich Rantzau was of the opinion that the only way to lay a solid foundation for astrology was the improvement of its astronomical requirements, i.e. more correct calculations of planetary positions, as he explicitly stated in a letter to Tycho at the beginning of 1587. In Rantzau's opinion Astronomy proper rests on solid foundations derived from observations, but also its conjectural part (i.e. astrology) is justifiable. Failures in predictions must not be ascribed to the art itself, but are the result of imperfect knowledge. Moreover, God is free to alter events prescribed in the course of the stars according to His own discretion. From Tycho's observatory and his instruments Rantzau expected advances in both areas of astronomy.¹³

Together with the rather casual remark on the book catalogue this is the only reference to a discussion of astrological matters between Rantzau and Tycho. But the correspondence is obviously incomplete. In 1594 Rantzau wrote to a certain Ludolph Riddershusen,¹⁴ that he had discussed with Tycho at length methods of directions and the problem whether the effect of precession should be taken into

7 „*Si catalogum Astrologicorum tuorum librorum ex amplissima tua Bibliotheca descriptum mihi transmittere non displicuerit, rem feceris longe gratissimam*“ (TBOO, vol. 7, p. 90).

8 TBOO, vol. 7, p. 124–127 (Uraniborg, 13.9.1588).

9 TBOO, vol. 7, p. 385–389 (Uraniborg, 21.12.1588).

10 N. A. Møller Nicolaisen, *Tycho Brahes Papirmølle paa Hven: Udgravningen 1933–34 og forsøg til rekonstruktion*, Copenhagen 1946.

11 TBOO, vol. 7, p. 299–301 (Uraniborg, 7.10.1590).

12 TBOO, vol. 8, p. 5f. (Segeberg, 17.9.1597); vol. 14, p. 124f. (Wandsbeck, 13.12.1597).

13 TBOO, Bd. 6, S. 29f. (Segeberg, 17.1.1587).

14 Riddershusen was born 1551 in Bremen (TBOO, vol. 8, p. 221), but apart from the date of his marriage and the date of birth of four children nothing is known about his life and profession. Research in the Bremen State Archive was in vain.

consideration when setting up astrological predictions.¹⁵ Later on Tycho himself was in contact with Riddershusen on house division and direction techniques.¹⁶ There is no comprehensive treatment of Tycho's astrological activities available.¹⁷ In his inaugural lecture *De disciplinis mathematicis* delivered at Copenhagen University in 1574 Tycho Brahe advocated the legitimacy both of natural and judicial astrology.¹⁸ As a court astronomer to King Frederik II. of Denmark Tycho was to deliver an annual almanac and he was commissioned to cast horoscopes for the newborn princes, Christian in 1577,¹⁹ Ulrik in 1579²⁰ and Hans in 1583.²¹ The horoscope for Prince Christian has not survived in original, but in two copies written after 1588, the year of his coronation. In the circular scheme customary used by Tycho a false date of birth is given; it should read April 12th. Tycho employed the Prutenic Tables for calculating the planetary positions and added a few corrections based on his own observations. Contrary to the majority of astrologers in the 16th century who used the „rational method“ preferred by Regiomontanus he employed a system of house division commonly ascribed to Campanus of Novara,²² i.e. positional circles joining in the north and south point

15 Schleswig, Landesarchiv Schleswig-Holstein: Abt. 127.21, Ms. 293, p. 1171–1173.

16 TBOO, vol. 8, p. 216–221 (Bremen, 16./26.12.1599); p. 304f. (Benatek, 10./20.4.1600).

17 Despite its promising title Franz Studnicka's publication Bericht über die astrologischen Studien des Reformators der beobachtenden Astronomie, Tycho Brahe: Weitere Beiträge zur bevorstehenden Saecularfeier der Erinnerung an sein vor 300 Jahren erfolgtes Ableben (Prague 1901) is limited to facsimiles of some astrological notes in books owned by Tycho accompanied by very poor commentaries.

18 TBOO, vol. 1, p. 143–173; for a summary of its contents see John Louis Emil Dreyer, *Tycho Brahe: A Picture of Scientific Life and Work in the Sixteenth Century*, Edinburgh 1890 (2nd ed. New York 1963), p. 74–78. There is also a German translation by Karl Zeller („Über die mathematischen Wissenschaften: Eine Rede Tycho Brahes (De Disciplinis Mathematicis)“, *Die Sterne: Monatsschrift über alle Gebiete der Himmelskunde*, 11, 1931, p. 100–122).

19 Horoscopus Regis Christian IV, ad mandatum Friderici II conscriptus in insula Hwena, a Tychone Brahe Uttonide, Cal. Jul. 1577 infanti Christiani inscriptus, nebst Astrolog. Urtheil von dieses jungen Herren Nativitet (Copenhagen, Det kongelige Bibliotek: Gamle kongelig Samling, 1821 4°; TBOO, vol. 1, p. 179–208; German translation by Søren Porsborg in: Lars Steen Michael, Erik Michael and Per Kjærgaard Rasmussen, *Astrologie: Von Babylon zur Urknall-Theorie*, Wien/Köln/Weimar 2000, p. 248–268). See also Dreyer (n. 18), p. 145–152.

20 Copenhagen, Det kongelige Bibliotek: Gamle kongelig Samling, 1822 4° (TBOO, vol. 1, p. 209–250).

21 Copenhagen, Det kongelige Bibliotek: Gamle kongelig Samling, 1823 4° (TBOO, vol. 1, p. 251–280).

22 The method was already employed in Iran and Central Asia in the 11th century and preferred by al-Biruni (973–after 1050), who claimed to be its inventor; see John David North, *Horoscopes and History* (= Warburg Institute: Surveys and Texts, vol. 13), London 1986, p. 29f., 32ff.; Edward Stewart Kennedy, „The Astrological Houses as Defined by Medieval Islamic Astronomers“, in: Josep Casulleras und Julio Samsó (Eds.), *From Baghdad to*

of the observer's horizon are laid at distances of 30° through the Prime Vertical, thus giving unequal sections of the ecliptic.

With the laborious task of calculating the positions of the planets and the cusps of the houses and examining the aspects – i.e. interpreting the radix horoscope – the astrologer's work was by no means finished. To foresee future events was most attractive. This was achieved by using methods of direction with the result of a distance measured along the celestial equator which had to be transformed into time.²³ Tycho's astrological interpretation of the geniture by using primary directions is divided into sections by topic: childhood, course of life, character, morals and disposition, body constitution, travels, marriage, children, friends and death. It concluded with the statement that no prediction is settled irrevocably but God disposes of everything according to His own will and judgment.

Tycho's book on the new star of 1572 also contains long astrological passages, in which the astrological significance of the event is discussed at length.²⁴ The horoscope Tycho cast for the event shows a system of eight houses which he obtained by dividing the Prime Vertical into steps of 45° .²⁵ By this time he obviously had completed an account on the various methods of house division, which was never published and is not extant in manuscript.²⁶ The same goes for a tract against contemporary astrological practice.²⁷

Tycho was keenly interested in Astrometeorology and from 1582 onwards he kept long records of the weather which were correlated to planetary positions, the appearance of comets and the moon's phases.²⁸ His work in this field is

Barcelona: Studies in the Islamic Exact Sciences in Honour of Prof. Juan Vernet, Barcelona 1996, vol. 2, p. 541.

23 The rather complicated technical procedures cannot be expanded here at length. No comprehensive historical treatment is available with the exception of a publication by Rüdiger Plantiko (*Primärdirektionen: Eine Darstellung ihrer Technik*, Mössingen 1996) which has to be read with some caution because of the author's obvious commitment to astrology.

24 *De Nova et Nullius Aevi Memoria Prius Visa Stella, iam pridem Anno a nato Christo 1572 mense Novembrij primum conspecta, Contemplatio Mathematica*, [...], TBOO, vol. 1, p. 35ff.

25 TBOO, vol. 1, p. 33.

26 *De variis astrologorum in coelestium domorum divisione opinionibus, earumque insufficiencia* (TBOO, vol. 1, p. 38f).

27 *Contra astrologos pro astrologia* (TBOO, vol. 1, p. 36).

28 See Peder Jacobsen Flemløse's *Astrologia* (Uraniborg 1591), a booklet with weather predictions with a preface by Tycho Brahe, although it runs under Flemløse's name (John Robert Christianson, „Tycho Brahe's Cosmology from the *Astrologia* of 1591“, *Isis: An International Review Devoted to the History of Science and its Cultural Influences*, 59, 1968, p. 312–318), and

characterized by a great concern for reliability. But in the course of time he got more and more sceptical mainly because of the insufficient precision to calculate planetary positions in advance. This is confirmed by a letter written in 1587 to Heinrich Below, Tycho's brother-in-law.²⁹ The Duke of Mecklenburg had obtained two prognostications issued by Tobias Möller and Andreas Rosa for the year 1588 of which one predicted a year governed by two beneficial planets, the other by two malevolent ones and requested Below to inquire which of them were correct. Tycho explained this fact by pointing out that one judgement was based on the Alfonsine Tables, the other on the Prutenic Tables. Neither of these tables were accurate enough as he had found by comparing the calculations derived from them with his own observations so Tycho concluded neither prediction could be relied upon. The date for Below's question was not accidental. Already in 1564 Cyprianus Leovitius (1524–1574), court mathematician to Count Palatine Ottheinrich,³⁰ had issued a booklet predicting horrible devastations, because in this very year (1588) twelve conjunctions in different signs of the zodiac and two occultations were to be expected.³¹ This very popular prediction sometimes was ascribed also to Johannes Stoeffler or Johannes Schoener and handed down in popular rhymes.³²

In his *Disputationes adversus astrologiam divinatricem* (1494) Pico della Mirandola had set out to demolish either the philosophical foundations of astrology and Ptolemy's astrological physics. He denied the antiquity of empirical records, i.e. he

the *Diarium astrologicum et meteorologicum anni a nato Christo 1586* (Uraniborg 1586) which Tycho published under the name of another pupil, Elias Olsen Morsing.

29 TBOO, vol. 7, pp. 116–119. The letter is printed also in: Georg Christian Friedrich Lisch, „Tycho Brahe und seine Verhältnisse zu Meklenburg“, *Jahrbücher des Vereins für mecklenburgische Geschichte und Alterthumskunde*, 34, 1869, p. 183–188, and Dreyer (n. 18), p. 384–386.

30 Günther Oestmann, „Cyprianus Leovitius, der Astronom und Astrologe Ottheinrichs“, in: Proceedings of a symposium on Count Palatine Ottheinrich, Neuburg an der Donau, Oct. 26th–28th, 2001 (*Pfalzgraf Ottheinrich: Politik, Kunst und Wissenschaft im 16. Jahrhundert*, Regensburg 2002, p. 348–359).

31 *De conjunctionibus magnis insignioribus superiorum planetarum, Solis defectionibus, et Cometis, in quarta Monarchia, cum eorundem effectuum historica expositione [...]*, Lauingen 1564; see Joseph Mayer, „Der Astronom Cyprianus Leovitius (1514–1574) und seine Schriften“, *Bibliotheca mathematica: Zeitschrift für Geschichte der mathematischen Wissenschaften*, Series III, 4, 1903, p. 148–150.

32 E.g. *Das tausent fünff hundert acht vnd achtzig jhar nimm war/ Geschicht nichts neues/ so vergehet die Welt gar*; see Ernst Zinner, *Leben und Wirken des Johannes Müller von Königsberg, genannt Regiomontanus* (= *Milliaria: Faksimiledrucke zur Dokumentation der Geistesentwicklung*, vol. 10.1), 2nd ed. Osnabrück 1968, p. 205f.; Robin Bruce Barnes, *Prophecy and Gnosis: Apocalypticism in the Wake of the Lutheran Reformation*, Stanford (Cal.) 1988, p. 163ff.

regarded the legendary observational tradition of the Chaldeans as fictitious. Moreover, Pico pointed out the inaccuracies of astronomical observations. Thus, astrologers in the 16th century were challenged to the solution of three problems: (i) to refine mathematical astronomy, (ii) the identification of a secure model of astrological physics, (iii) the necessity of defining reliable textual authority, and of establishing the correct reading of those texts.³³ Viewed in this broader context, Tycho Brahe's and Heinrich Rantzau's attitude towards astrology may be regarded as complementary. Concern for empirical reliability was characteristic for the astrological activities of both men, but while Tycho attempted to fulfil the first part of this „research program“ by most accurate observations and his physico-astrological ambitions in meteorology, Heinrich Rantzau was engaged in publishing compilations and manuscript sources with the aim of establishing a secure foundation to the art.

Finally in his description of the instruments Tycho employed on the island of Hven (*Astronomiae Instauratae Mechanica*, Wandsbeck 1598) he rendered account to the reader for his intentions and achievements. Concerning astrology he set out:

„In the field of Astrology, too, we carried out work that should not be looked down upon by those who study the influences of the stars. Our purpose was to rid this field of mistakes and superstition, and to obtain the best possible agreement with the experience on which it is based. For I think that it will hardly be possible to find in this field a perfectly accurate theory that can come up to mathematical and astronomical truth. Having in my youth been more interested in this foretelling part of Astronomy that deals with prophesying and builds on conjectures, I later on, feeling that the courses of the stars upon which it builds were insufficiently known, put it aside until I should have remedied this want. After I at length obtained more accurate knowledge of the orbits of the celestial bodies, I took Astrology up again from time to time, and I arrived at the conclusion that this science, although it is considered idle and meaningless not only by laymen but also by most scholars, among which are even several astronomers, is really more reliable than one would think; and this is true not only with regard to meteorological influences and predictions of the weather [natural astrology], but also concerning the predictions by nativities [judicial astrology], provided that the times are determined correctly, and that the courses of the stars and their entrances into definite sections of the sky are utilized in accordance with the actual sky, and that their directions of motion and revolutions are correctly worked up. With regard to these points we have developed a method, based on experience, which differs from those used

33 As pointed out by Steven Vanden Broecke, *The Limits of Influence: Astrology at Louvain University 1520–1580*, PhD thesis Leuven 2000, p. 79, who examined the attempts for an astrological reform at Louvain.

up to now. But we are not inclined to communicate this kind of astrological knowledge to others, since not a little has been made out by us in this field. For it is not given to everybody to know how to use it on their own, without superstition or excessive confidence, which is not wise to show towards created things. Therefore we shall not publish any, or at least very little, of the things that we have found out in this field.“³⁴

Thus Tycho’s astrological ambitions were more than an aberration supposed to be characteristic for an astronomer of the sixteenth century. His aim was the construction of a reformed astrology, but for a publication he thought time not yet ripe. Astrology and Alchemy were integral components of Tycho’s cosmology and this view found an artistic expression in two vignettes for his books *De mundi aetherei phaenomenis* (1588), *Epistolarum astronomicarum* (1596) und *Astronomiae instauratae mechanica* (1598). Both vignettes show a man in reclining posture and a youth, but in one case the man is looking upward into the sky and leaning on a globe with a pair of compasses in his hands. In the other case he is accompanied by a chemical apparatus, holds a bunch of herbs in his hands and looks downward. The vignettes bear the meaningful captions *SUSPICIENDO DESPICIO* (By looking up I see downward) and *DESPICIENDO SUSPICIO* (By looking down I see upward).

34 Hans Raeder, Elis and Bengt Strömgren (Eds. and transl.), *Tycho Brahe’s description of his instruments and scientific works as given in his Astronomiae Instauratae Mechanica (Wandesburgi 1598)*, Copenhagen 1946, p. 117f. Latin text in TBOO, vol. 5, p. 117: „*In Astrologicis quoque effectus siderum scrutantibus non contemnendam locavimus operam, ut et haec, a mendis et superstitionibus vindicata, experientiae, cui innituntur, ut plurimum consona sint. Nam exactissimam in iis adinvenire rationem, quae Geometricae et Astronomicae veritati par sit, minus duco possibile. Cum vero huic Prognosticae Astronomiae parti, quae mantica et Stochastica est, in adolescentiâ impensius addictus fuisset, posteaque ob motus Siderum, quibus fundatur, non satis perspectos eam seposuisset, donec huic incommodo subveniretur; compertis demum exactius Siderum viis, eam subinde in manus resumendo, majorem subesse certitudinem huic cognitioni, utut vana et frustranea non solum vulgo, sed et plerisque Doctis, adeoque nonnullis inter eos Mathematicis habeatur, comperi, quam quis facile existimârit: Idque tam in influentiis et praedictionibus meteorologicis, quam Genethliacis, modo tempora rite constant, et motus Siderum atque ingressus Coelo consoni adhibeantur, ac directiones atque revolutiones rite administrentur: In quibus duobus nos etiam aliam ab ipsâ experienciâ extruximus rationem, quam hactenus usitatum fuit. Sed nos istiuscemodi Astrologica non libenter aliis impertimur, quatenus haud pauca in his explorata habemus: siquidem non omnes eâ quâ decet circumspectione citra superstitionem et nimiam confidentiam, quae nullis creaturis tribuenda est, discrete uti nōrint. Ideoque aut nulla aut admodum pauca ex nostris inventis de his in publicum evulgabimus.*“